



Multi-type data fusion framework based on deep reinforcement learning for algorithmic trading

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Abstract

In recent years, research on algorithmic trading based on machine learning has been increasing. One challenge faced is getting an accurate representation of the stock market environment from multi-type data. Most existing algorithmic trading studies analyze the stock market based on a relatively single data source. However, with the complicated stock market environment, different types of data reflect the changes in the stock market from different perspectives, and how to obtain the temporal features of different types of data and integrate them to obtain a deeper representation of the stock market environment are still problems to be solved. To tackle these problems, in this study, we combine deep learning and reinforcement learning (RL) and propose a multi-type data fusion framework with deep reinforcement learning (MSF-DRL) that integrates stock data, technical indicators and candlestick charts, in which technical indicators can reduce the impact of noise in stock data. In the process of learning trading strategies under the MSF-DRL framework, the temporal features of stock data and technical indicators are extracted through a long short-term memory (LSTM) network, and a convolutional neural network (CNN) and bidirectional long short-term memory (BiLSTM) are successively used to extract the features of the candlestick chart. The fused features are used as the input of the RL module, which makes trading decisions on this basis. To verify the effectiveness of the MSF-DRL framework, we conducted comparative experiments on datasets composed of Chinese stocks and some stocks of the S&P 500 stock market index. Compared with the other trading strategies, our trading strategy can obtain more profits and a higher Sharpe ratio.

Keywords Deep reinforcement Learning · Algorithmic trading · Multi-type data · Trading strategy

1 Introduction

In recent years, the stock market has attracted an increasing number of investors, and the purpose of investors is to maximize profit. However, due to the influence of external factors, such as natural disasters and emergencies, the price of stocks fluctuates greatly and is nonstationary [1–4]. The key to stock trading is to develop a suitable trading strategy; that is, to make the right decisions at the right times, a good trading strategy yields more profits and stops losses in time. In general, investors make trading decisions based on their own judgments on the stock market, however, they

are often affected by their own sentiment in the process of trading, which results in the final profits often falling short of expectations.

Recently, machine learning methods have been increasingly used in stock trading. The process of replacing investors with algorithmic models for automated trading is called algorithmic trading [5] and eliminates the influence of investor sentiment on the entire trading process. Algorithmic trading can be considered a decision-making process, and the challenge it faces is continuously perceiving the environment and making optimal trading decisions. Reinforcement learning (RL) [6] is learning a mapping from the environment state to the action, which solves the problem of sequential decision making. RL has been successful in game playing [7, 8], robotic applications [9, 10], etc. When algorithmic trading is implemented based on RL, dynamic trading strategies can be learned by interacting with the stock market environment. Chakole et al. [11] learned a

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dynamic trading strategy based on the Q-learning algorithm. However, in algorithmic trading, although RL can learn independently and make continuous decisions, it lacks the ability to perceive the stock market environment. Deep learning [12] can extract abstract feature representations from original data through a model composed of multilayer linear units and has strong perception abilities; Sezer et al. [13] made full use of the advantages of a convolutional neural network (CNN) to analyze the two-dimensional (2D) images converted from financial data and make trading decisions, but the trading strategy based on deep learning cannot be adjusted in time according to the changes in the stock market. Deep reinforcement learning (DRL) combines the perception ability of deep learning and the decision-making ability of RL, and it is more advantageous to implement algorithmic trading based on DRL.

Currently, research on algorithmic trading based on DRL is in the development stage, and there are algorithmic trading studies that combine deep neural network (DNN) to construct a DRL model. Deng et al. [14] combined fuzzy learning, DNN and recurrent neural network (RNN) to implement algorithmic trading and proposed a task-aware propagation through time method to solve the problem of gradient disappearance during network training. Théate et al. [15] framed the algorithmic trading decision-making problem as an RL problem, and constructed the agent in RL through a DNN to implement the deep Q-network (DQN) algorithm to solve the problem. Although these methods implement algorithmic trading by autonomously making trading decisions by perceiving the state of the environment, they ignore the importance of the temporal relationship between financial data.

To better summarize the state of the environment, the algorithmic trading studies based on DRL pay more attention to obtaining the temporal dependence between financial data. Lei et al. [16] proposed the time-driven feature-aware jointly DRL (TFJ-DRL) model, analyzed the stock data and technical indicators by a gated recurrent unit (GRU), and introduced an attention mechanism to summarize the features of the stock market environment. Lee et al. [17] first used the wavelet transform to remove noise in the stock data, and then the agent constructed by long short-term memory (LSTM) analyzes the data after noise removal and makes trading decisions. Ma et al. [18] implemented the advantages of fully connected (FC) and LSTM modules to analyze the current state of stocks and long-term historical trends to obtain the features of the stock market environment and make trading decisions on the basis. Although these studies summarized the environment state by capturing the temporal features of financial data, the data source analysis is relatively simple. The 2D image candlestick chart converted

from stock data can reflect the stock market information in different modes. Through quantitative analysis, we found that the temporal analysis of multi-source data could better summarize the stock market environment.

In this paper, multi-source data refers to stock data, technical indicators, and candlestick charts, which are essentially different types of data. Multi-type financial data contains abundant information, aiming at the problem of how to summarize the environment state from multi-type financial data, we propose a multi-type data fusion framework with DRL (MSF-DRL) that fuses the temporal features of the stock market environment reflected by stock data, technical indicators and candlestick charts. The fusion of differently structured data sources facilitates the learning of repeated features and different features of each data source, obtaining a more accurate feature representation of the stock market environment, which is more conducive to learning the optimal trading strategy and thus more benefits. In the MSF-DRL framework, deep learning is used to perceive the stock market environment, RL makes trading decisions on this basis, and RL module combines the Double DQN and Dueling DQN. Regarding the problem of multi-type data fusion and considering the different structures of data sources, we use different deep neural networks to extract their temporal features, and add the obtained features to achieve fusion.

In the experiment, we compare our trading strategy learned based on the MSF-DRL framework with other trading strategies on practical trading data and evaluate the results of the experiment through evaluation indicators such as the average annualized return (AR), Sharpe ratio (SR), and maximum drawdown (MDD). For stocks with different price trends in different fields, the experimental results show that compared with the trading strategies learned by relatively single data, our trading strategies learned by multi-type data can obtain higher profits and SR than other trading strategies in the same situation, showing a good generalization ability. In terms of feature fusion of multi-type data, for stocks with different trends, we implement feature fusion based on addition, concatenation and bilinear pooling. Experimental results show that the trading strategy learned from feature fusion of multi-type data through addition can obtain higher returns and SR. In addition, we conduct ablation experiments to verify the effectiveness of the selected data types and further conclude that analysis and fusion of multi-type data can preferably summarize the environmental state and learn better trading strategies.

The main contributions of this research are as follows:

1. A multi-type data fusion framework MSF-DRL based on DRL is proposed for algorithmic trading. The

MSF-DRL framework integrates the information of stock data, technical indicators and candlestick charts, and the experimental results prove that for stocks with different trends in different fields, the trading strategies learned under the MSF-DRL framework show better generalization ability.

2. Compared with two different fusion methods of concatenation and bilinear pooling, the MSF-DRL framework achieves fusion by adding different data source features, which is more helpful for learning the optimal trading strategy.
3. Taking into account the difference in the structure of the data sources, a LSTM is used to extract features of stock data and technical indicators, and a CNN and bidirectional long short-term memory (BiLSTM) are successively used to extract the temporal features of candlestick charts.

The rest of the paper is organized as follows: Section 2 generally shows related work. Section 3 provides the details and background of the MSF-DRL framework. Section 4 mainly describes the details of the experimental implementation. Section 5 discusses the experimental results, and the conclusion and future work of this paper are described in Section 6.

2 Related work

At present, some achievements have been made in research on algorithmic trading, and an increasing number of studies have implemented algorithmic trading based on machine learning methods. Algorithmic trading based on machine learning methods can directly learn trading strategies from information related to stocks. Compared with traditional trading strategies, trading strategies learned based on machine learning methods have more advantages because they can discover potentially beneficial trading patterns.

In the face of a complicated stock market, some studies implement algorithmic trading based on deep learning. Deep learning can extract informative features from different categories of raw data and has achieved significant success in many fields, such as computer vision [19–21], natural language processing [22–24], and speech recognition [25–27]. In the financial field, many studies implement algorithmic trading by predicting stock prices or trends in the future. Fischer et al. [28] used LSTM to forecast the direction of stock trends to make trading decisions. Beak et al. [29] proposed the ModAugNet-c model to predict the stock market index; the ModAugNet-c model is composed of an overfitting prevention LSTM module and a prediction

LSTM module and takes the target stock and relevant stocks combinations as input. Zhou et al. [30] combined empirical mode decomposition (EMD) and factorization machine-based neural network (FNN) to propose a 2-stage, end-to-end model EMD2FNN for stock market trend prediction. However, the trading strategies learned based on deep learning methods are often static. In the face of a volatile stock market, compared with static trading strategies, dynamic trading strategies can adjust trading decisions according to changes in the stock market, which has more advantages.

RL is used for algorithmic trading due to its ability to autonomously learn and make decisions. RL can be used to solve the Markov decision problem [31]. The agent in RL is a component of making action choices to obtain more accumulated rewards. When using RL to implement algorithmic trading, the agent is equivalent to the role of an investor and continuously interacts with the stock market environment to gain more profits. For example, Chakole et al. [11] used the Q-learning algorithm to find the optimal trading strategy and proposed two models to represent the state of the environment with clusters and candlestick charts. Wu et al. [32] proposed a portfolio management system based on RL that combines a CNN and a RNN to configure assets through the analysis of financial data. Although algorithmic trading based on RL can learn dynamic trading strategies, RL lacks the ability to perceive the environment in the face of a complicated stock market.

Recently, more research based on DRL for implementing algorithmic trading has been performed. Deng et al. [14] combined deep learning and RL to implement online financial assets trading, and fuzzy learning was introduced to reduce the uncertainty of the input data. Wu et al. [33] proposed two trading strategies, namely, the gated deep Q-learning trading strategy (GDQN) and the gated deterministic policy gradient trading strategy (GDPG) by the combination of a GRU with the Q-learning algorithm and policy gradient algorithm, respectively, and relevant technical indicators were used to reduce the impact of noise in stock data. Carta et al. [34] proposed a multi-layer and multi-ensemble approach, which makes full uses of the advantages of deep learning and DRL, and use multiple integrated steps to provide its final intra-day trading strategy. In the stock market, technical indicators are calculated based on historical stock data and thus include extensive important information and help in the analysis of nonlinear and unstable stock markets [35].

Most of the research on algorithmic trading based on DRL analyzes the stock market conditions based on relative single data source. In this research, we propose a multi-type data fusion MSF-DRL framework based on

deep DRL that fuses the temporal features of stock data, technical indicators and candlestick charts. For the purpose of extracting the features of different data sources, the LSTM network is used to extract the temporal features of stock data and technical indicators. Taking into account the different structures of the candlestick chart, we use a CNN and BiLSTM to extract the temporal features. The fused features can better reflect the state of the stock market than can a single feature and are more helpful for learning the optimal trading strategy.

3 Methodology

We combine deep learning and RL and propose the multi-type data fusion MSF-DRL framework for algorithmic trading. In this section, first, the overall MSF-DRL framework is introduced. Second, the deep learning perception module is described in detail. Finally, RL is specifically applied to algorithmic trading.

3.1 Overall structure of the framework

The main purpose of investing in the stock market is to obtain investment profits. When stock trading based on DRL is implemented, correctly perceiving the environment of the stock market and making the correct decision in time are key components. In order to reflect the stock market environment state at a deeper level and learning the optimal trading strategy, we propose the MSF-DRL framework, which integrates the information of stock data, technical

indicators and candlestick charts. Fig. 1 shows the overall structure of the framework.

Figure 1 clearly illustrates the position of each module in the overall MSF-DRL framework, which can be divided into two modules, the deep learning perception module and the trading decision module based on RL. The deep learning perception module in this research extracts corresponding features from stock data, technical indicators and candlestick charts to better capture the stock market state. The candlestick charts are transformed from stock data, and their features are extracted by the CNN and BiLSTM. The CNN is composed of a convolutional layer and a subsampling layer, and it can learn multiple layer features in the hierarchical structure. However, a CNN cannot capture temporal features, whereas a BiLSTM can model the bidirectional time structure and combine the current state and the next state to address the long-term dependency of the data. Therefore, we obtained the features of the candlestick chart through a CNN and a BiLSTM successively. An LSTM is a variant of an RNN that contains forget gates, output gates, and input gates, and the complexity of the LSTM network avoids the problems of gradient disappearance and gradient explosion and enables capturing temporal features of data. Thus historical stock data are cleaned, and technical indicators are calculated on this basis; moreover, both are used as the input of the LSTM network for feature extraction. In the module of trading decisions based on RL, the features extracted by the deep learning module are used as the environmental state, and the agent in the RL makes trading decisions on this basis. In other words, the continuous state series can simulate changes in the stock

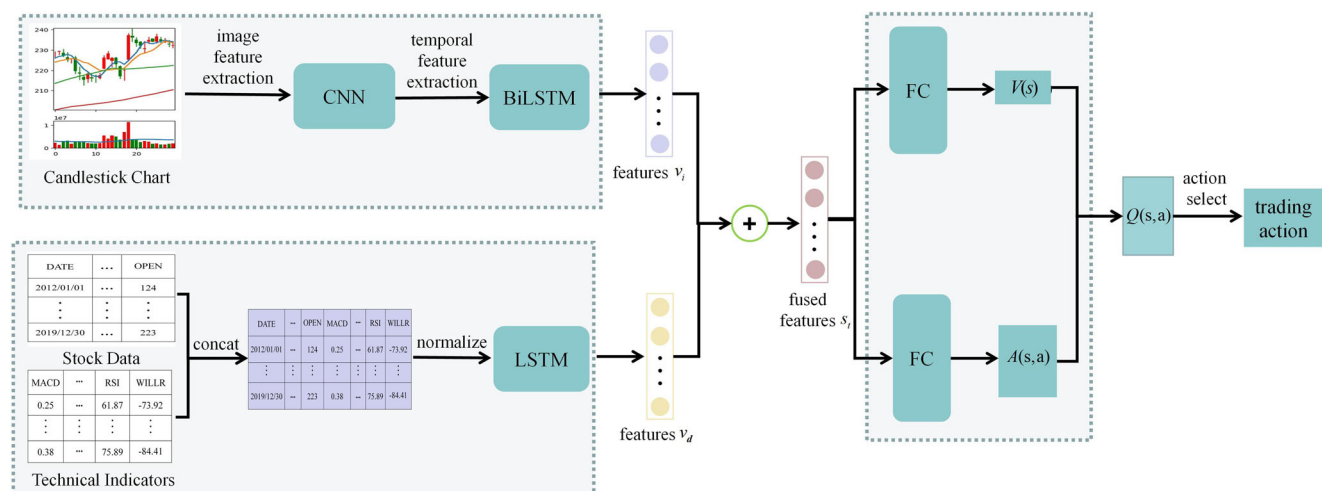


Fig. 1 Overall structure of framework MSF-DRL (The candlestick chart first through CNN to obtain abstract image features, and then inputs the features as sequence data to BiLSTM to obtain temporal features v_i . After the stock data and technical indicators are concatenated and normalized, they are used as the input of LSTM to obtain temporal features v_d . The feature vector obtained by adding these two feature

vectors is used as the input of the two FC layers. In this paper, we use the FC layers to construct the Dueling DQN network, the two FC layers represent the advantage function $A(s, a)$ and state value function $V(s)$ in Dueling DQN, the final Q value is obtained by adding the outputs of the two functions)

market environment, so the agent can constantly interact with the stock market environment and learn the optimal trading strategy through a constant process of trial and error. In addition, the DQN algorithm is a combination of deep learning and Q-learning, and since there is a problem of overestimating the Q value, a combination of the Double DQN and Dueling DQN is adopted in the RL module to avoid the overestimation of the Q value and improve the convergence speed of the model.

In this study, DRL combines the advantages of deep learning and RL, and different deep neural networks are used to extract and integrate the temporal features of different data sources, which is more conducive to learning the optimal dynamic trading strategy.

3.2 Deep learning perception module

This study aims to obtain a deeper feature representation of the environment of the stock market through the fusion of multi-type data to learn the optimal trading strategy. Although raw stock data can reflect changes in the stock market, they contain considerable noise. To reduce the effect of noise and to more objectively and accurately perceive the stock market changes, this research adds relevant technical indicators based on raw stock data. In summary, the input of the entire framework includes stock data, technical indicators and candlestick charts.

3.2.1 Stock data and technical indicator feature extraction

Since stock data contain considerable noise, we use relevant technical indicators to reduce the impact of noise, and technical indicators can reflect changes in the stock market from different perspectives. In this paper, stock data and technical indicators are used as inputs of the LSTM network to better capture the main trends of the stock market. The raw stock data we use include opening price, closing price, high price, low price and trading volume. The technical indicators used in this research are the MACD, EMA, DIFF, DEA, KDJ, BIAS6, BIAS12, BIAS24, RSI6, RSI12, RSI24 and WILLR; the indicators are calculated by mathematical formulas based on stock prices and trading volumes, as reported in Table 1.

To facilitate subsequent calculations, we perform missing value processing on stock data. We clean the stock data first, and the missing data are supplemented with 0. In addition, the input of the neural network must be a real value, so we replace the data types of not a number (NaN) in stock data and technical indicators with 0. Due to the wide range of stock prices, trading volumes and technical indicators, data with different value ranges may experience gradient explosion in the process of neural network training. To prevent this problem, we carry out normalization processing

Table 1 The list of used technical indicators

Technical indicator	Indicator description
MACD	Moving Average Convergence / Divergence
EMA	Exponential Moving Average
DIFF	Diff
DEA	Difference Exponential Average
KDJ	Stochastics
BIAS	Bias
RSI	Relative Strength Index
WILLR	Williams % R

on the stock data and technical indicators, and normalization is performed to transform the data to a fixed interval. In this work, the stock data and technical indicators of each dimension are normalized, and the data are converted to a range [0, 1]. The normalization formula is as follows:

$$X_{\text{norm}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad (1)$$

where X represents the original data, X_{norm} represents the normalized data, $X_{\min}$ and $X_{\max}$ represent the minimum and maximum values of the original data, respectively.

The neural network structure for extracting stock data and technical indicators is shown in Fig. 2. In the LSTM network, f , i and o represent a forget gate, an input gate and an output gate, respectively; C_t is the state of the memory cell at time t ; $\tilde{C}_t$ is the value of the candidate state of the memory cell at time t ; σ and $\tanh$ are the sigmoid and tanh activation functions, respectively; W and b represent the weight and deviation matrix, respectively; x_t is the input vector; h_t is the output vector; x_t and other specific calculation formulas are as follows:

$$x_t = \{ \text{open, close, } \dots, \text{MACD, RSI, BIAS} \} \quad (2)$$

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (3)$$

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (5)$$

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (6)$$

$$o_t = \sigma(W_o \cdot [h_{t-1}, x_t] + b_o) \quad (7)$$

$$h_t = o_t * \tanh(C_t) \quad (8)$$

In the whole process of stock data and technical indicator feature extraction, the stock data and technical indicators are first cleaned and normalized. Then, as the input of the LSTM network, at least two circulation layers are generally required when processing sequential data [36]; considering the complexity of the network, we adopt a two-layer LSTM network, where the second LSTM layer can update its internal state according to the content obtained by the first

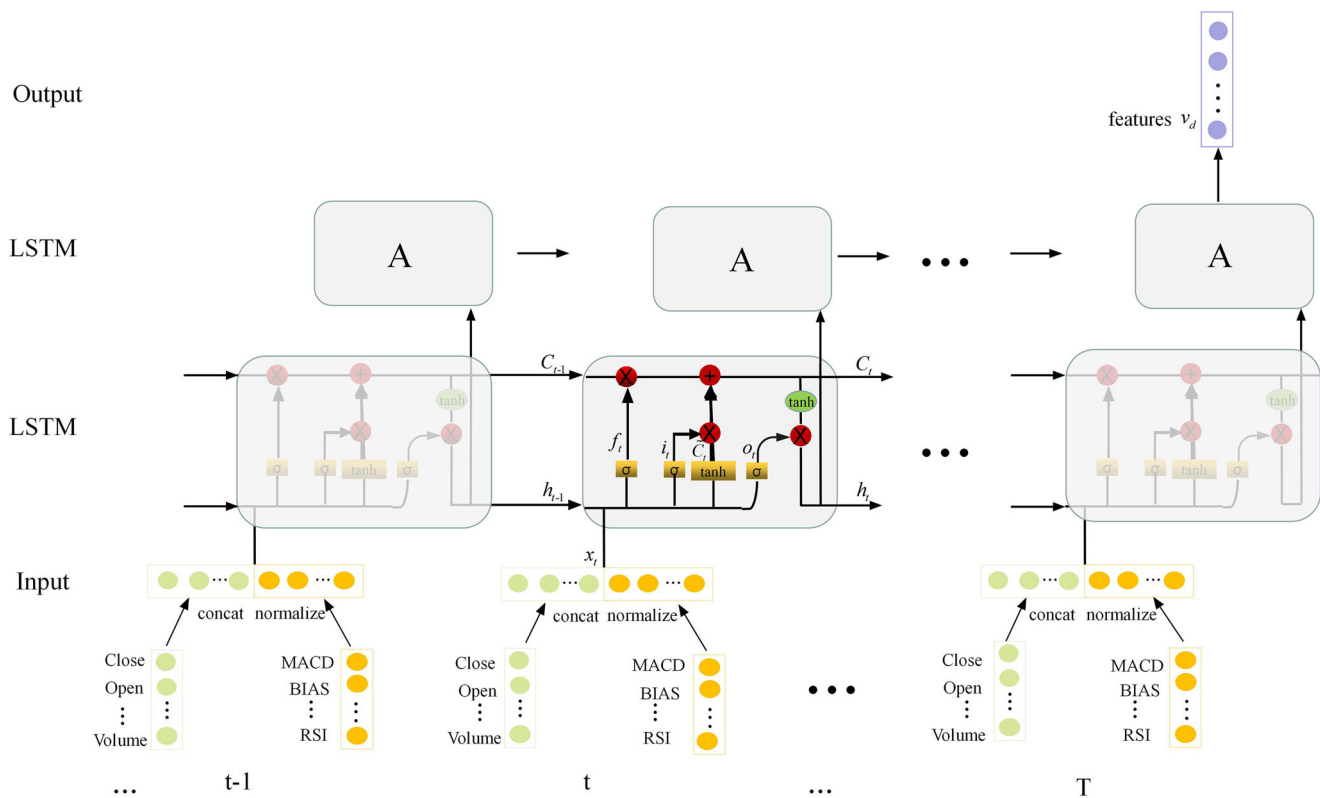


Fig. 2 Network structure for extracting data features (A is the LSTM network structure unit, x_t and h_t are the input vector and output vector at time t , respectively. f , i and o are forget gate, input gate and output

gate, respectively. C_t is the state of the memory cell at time t , $\tilde{C}_t$ is the value of the candidate state of the memory cell at time t)

LSTM layer to capture long-term dependency, and the final features are obtained through the two-layer LSTM network.

3.2.2 Image data feature extraction

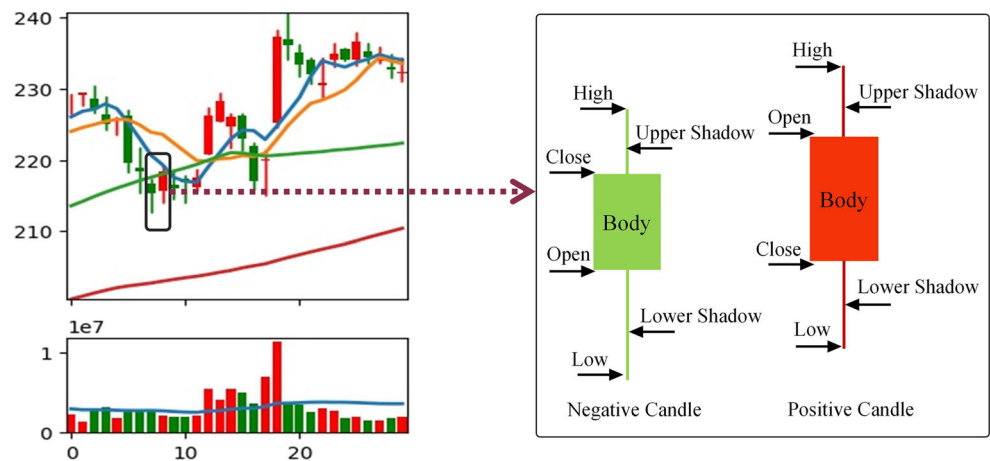
To extract more informative features, in this study, based on the K-line theory, historical stock data are transformed to candlestick charts. In the process of transformation, the historical stock data (high, open, low, close, volume) are first processed by K-line processing, and then the data are calculated to obtain the length of the upper shadow line, the lower shadow line, the body, the color of the K-line body, the moving average of the close price and the trading volume; Among them, we use the technical analysis library (TA-Lib)¹ to calculate the closing price moving average and the trading volume moving average. In addition, in the histogram of trading volume, the color of the trading volume of the day is the same as the color of the K-line body, so

these charts contain not only the candlestick but also other information and can be divided into two parts, as shown in Fig. 3. The upper part is the candlestick and 5-day, 10-day, 60-day, 120-day moving averages of the closing price; the lower part is the trading volume histogram and its moving average. Generally, in the candlestick, when the color of the candlestick is green, it represents the negative candle and indicates that the opening price of the stock is higher than the closing price of the day and that the stock price falls. When the color of the candlestick is red, it represents the positive candle and indicates that the closing price of the stock is higher than the opening price of the day and that the stock price rises. The candlestick chart can reflect the overall trend of stock prices, where the trading time of stocks can range from one minute to one month, the candlestick charts in this study use days as the unit.

The network structure of extracting image data features is shown in Fig. 4, and since the candlestick chart transformed from stock data, it still needs a neural network to extract its temporal features. In this work, first, CNN is used to extract candlestick chart features, which contains three-layer convolution and pooling. CNN can learn multiple

¹<https://ta-lib.org/>

Fig. 3 Candlestick chart (The candlestick chart shows the changes in stock price and trading volume during the trading period)



layers features in the hierarchy, nonetheless, it cannot handle the temporal dependence among time series data. Therefore, the feature vectors extracted by CNN are input to BiLSTM as sequence data to obtain the temporal features of the candlestick chart. The BiLSTM is composed of the forward LSTM and the backward LSTM. Since the features obtained after the image data passes through the CNN are becoming increasingly abstract, we use the BiLSTM network to further obtain the temporal features by connecting the correlations before and after the features.

3.3 Trading decision module based on reinforcement learning

In RL, the agent chooses an action according to the environmental state and obtains an immediate reward every time. The agent constantly adjusts the learning strategy according to the reward value to obtain the largest cumulative reward. For example, in the process of stock trading, if the trading action selected by the agent gains a profit, the agent obtains a positive reward value; in contrast, if there is a loss after a trading action is chosen, the agent obtains a negative reward value. The reward value promotes the agent to take the correct action in future behavior choices. In this research,

we set the immediate reward based on the change in the asset value after each transaction decision made by the agent. The immediate reward r and cumulative rewards R_t in this study are calculated as follows:

$$r = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (9)$$

$$R_t = \sum_{k=0}^T \gamma^k r_{t+k+1} \quad (10)$$

where P_t is the sum of the assets owned by the investors at time t , $\gamma \in (0, 1)$, which is a discount factor.

DRL can be divided into two categories: value function-based DRL and strategy gradient-based DRL. In the value-based DRL algorithm, each action has a corresponding Q value, and the agent chooses the action according to the Q value under the ε -greedy strategy. For example, given the probability $\varepsilon = 0.8$, the agent selects the action corresponding to the maximum Q value at 80% probability and randomly selects the action with a probability of 20%.

The DQN is a value-based DRL algorithm. In the DQN algorithm, there are two networks with the same structure, the main network and the target network; initially, the

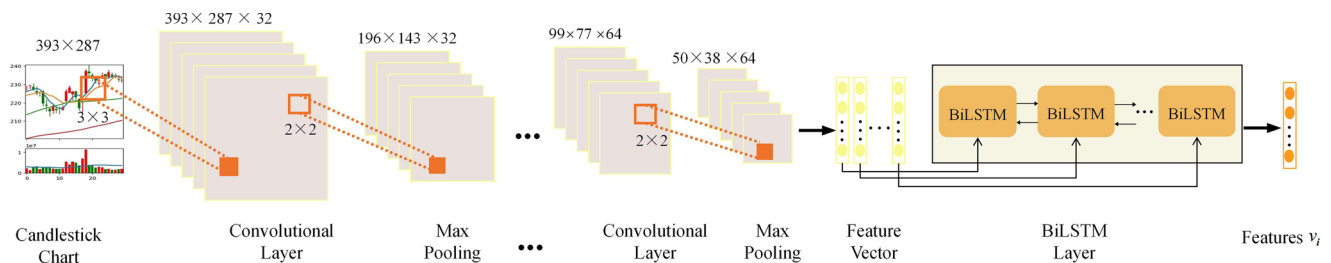


Fig. 4 Network structure for extracting image features

parameters of the two networks are exactly the same. In addition, the experience replay mechanism is adopted to reduce the strong correlation between data. In the training process, the mean square error of the two Q values of the main network and the target network is used as a loss function to train and update the parameters of the main network, and the parameters of the main network are assigned to the target network every fixed number of times. However, the DQN overestimates the Q value. The Double DQN and Dueling DQN are improved algorithms based on DQN, and both adopt different ways to improve Q value overestimation in the DQN.

In the Dueling DQN algorithm, the Q value is obtained by adding the state value function $V(s)$ and advantage function $A(s, a)$. In this study, $V(s)$ and $A(s, a)$ are represented by two fully connected layers. In stock trading, $V(s)$ represents the potential value of the current state of the stock market, while $A(s, a)$ represents the additional value brought to the investment income after action a is taken in the current state s . The calculation formula of the Q value is as follows:

$$Q(s, a; \theta, \alpha, \beta) = A(s, a; \theta, \alpha) + V(s; \theta, \beta) \quad (11)$$

where α and β represent the parameters in the two fully connected layers and θ represents other parameters in the MSF-DRL framework.

The Double DQN also solves the problem of value function overestimation and can be combined with the Dueling DQN to accelerate the convergence of the model. The Double DQN calculates the Q value of the target network in a different way than the DQN. The specific formula for calculating the Q value of the target network at time t is as follows:

$$Y_t = r_{t+1} + \gamma Q \left(s_{t+1}, \arg \max_a Q(s_{t+1}; \theta_t); \theta'_t \right) \quad (12)$$

where θ and θ' represent the parameters in the main network and target network, respectively.

The loss function in this study still uses the loss function of the DQN algorithm, and the specific formula is as follows:

$$L(\theta) = E \left[(Y_t - Q(s, a; \theta))^2 \right] \quad (13)$$

Stock trading as a continuous process can occur at any time. In this study, the trading frequency is in units of the days, and the trading price is the closing price of the day. The state of the stock market can be obtained by the addition of stock data features, technical indicator features and image data features extracted by the deep learning perception module :

$$s_t = v_i + v_d \quad (14)$$

where s_t is the state at a certain moment t , v_i is the image data features, and v_d is the features of the stock data and technical indicators. There are only two options for trading actions in the trading process, trading action $a_t = \{ \text{long, short} \} = \{1, -1\}$. For stocks in the Chinese stock market, the initial capital is 100,000 Chinese Yuan (CNY), for stocks in the S&P 500 stock market, the initial capital is 100,000 USD. Considering that in the process of trading, transaction costs are inevitable and affect the final trading results, especially when the cost is relatively high, the cost of transaction is higher. In addition, the transaction cost setting can prevent high-frequency trading. Therefore, this research takes the transaction cost into consideration in the process of implementing algorithmic trading.

Algorithm 1

Input: Stock data, technical indicators, candlestick chart;

Output: Trading strategy;

Initialize the experience replay memory D to capacity C ;

Initialize the main Q network with random weights θ ;

Initialize the target Q network with $\theta' = \theta$;

for episode 1 to N **do**

for $t = 1$ to T **do**

 The features of the candlestick chart obtained by CNN are input as sequence data into BiLSTM to obtain the temporal features v_i ;

 Stock data and technical indicators through LSTM network to obtain feature v_d ;

 The sum of the features extracted by deep learning module represents the environment state s_t ;

$s_t = v_i + v_d$;

 Calculate $A(s_t, a)$, $V(s_t)$ according to s_t ;

 Add $A(s_t, a)$ and $V(s_t)$ to get the Q value of each trading action;

$Q(s_t, a) = A(s_t, a) + V(s_t)$;

 With the probability ε choose a random action a_t ;

 Otherwise select $a_t = \arg \max_a Q(s_t, a; \theta)$;

 Get the reward r_t and next state s_{t+1} ;

 Store the transaction (s_t, a_t, r_t, s_{t+1}) to D ;

if $t \% n = 0$ **then**

 Sample minibatch (s_t, a_t, r_t, s_{t+1}) randomly from D ;

 Set:

$$Y_t = \begin{cases} r_t & \text{if the state } s_{t+1} \text{ is terminal,} \\ r_t + \gamma Q \left(s_{t+1}, \arg \max_a Q(s_{t+1}; \theta_t); \theta'_t \right) & \text{otherwise.} \end{cases}$$

 Train the network with loss function

$$L(\theta) = E \left[(Y_t - Q(s, a; \theta))^2 \right]$$

 Update the target network parameters $\theta = \theta'$ every N steps;

end if

end for

end for

The algorithmic trading pseudocode implemented by the MSF-DRL framework is shown in Algorithm 1. The inputs of the algorithm are stock data, technical indicators and a candlestick chart, and the output is the final learned trading

strategy. First, the network parameters are initialized. Second, the candlestick chart is first extracted through the CNN, and then the obtained feature sequence is taken as the input of BiLSTM to obtain feature v_i . Stock data and technical indicators are concatenated and normalized as the input of the LSTM network to get feature v_d . v_i and v_d are added together as the environment state s_t . Finally, $A(s_t, a)$ and $V(s_t)$ are calculated based on s_t , the two are added to obtain the Q value, and trading decisions are made according to the ε -greedy strategy. In the training process, Double DQN is used to calculate the Q value of the target network, and network parameters are optimized through the mean square error of the Q value of the target network and the main network to learn the optimal trading strategy.

The time complexity determines the training time of the model. The time complexity of CNN is Time $\sim O(M^2 \times K^2 \times \text{Cin} \times \text{Cout})$, M is the size of the output feature map, K is the size of the filter, Cin is the number of input channels, and Cout is the number of output channels. The time complexity of BiLSTM is Time $\sim O(M^2 \times K^2 \times 2\text{Cin} \times 2\text{Cout})$. The time complexity of LSTM is Time $\sim O(N \times D^2)$. N is the dimension of the vector, D is the sequence length. The time complexity of the DQN algorithm is Time $\sim O(P \times Q)$. P is the number of time steps in each episode, Q is the number of episodes. The time complexity of Algorithm 1 is Time $\sim O(P \times Q \times (M^2 \times K^2 \times 2\text{Cin} \times 2\text{Cout}))$. Although compared with the traditional algorithmic trading model, the time complexity of Algorithm 1 is higher, but compared with the algorithmic trading model based on machine learning, the time complexity of Algorithm 1 is the same as these models in order of magnitude.

Space complexity determines the number of model parameters. CNN can obtain the candlestick chart features with rich information, and input the features into BiLSTM to capture the temporal dependence. In the algorithm, the dropout mechanism is introduced in the fully connected layer, and part of the training parameters are randomly discarded in each iteration, which effectively prevent overfitting, and reduces the space complexity to a certain extent.

CNN has a wide range of applications in image data and has yielded good results [37]. CNN can obtain the spatial features of the candlestick chart through convolution and pooling, BiLSTM can capture bidirectional dependencies between sequential data [38], and the spatial features of the candlestick chart can be input into BiLSTM to further obtain the temporal features. LSTM solves the problem of gradient disappearance and gradient explosion in RNN and can capture the temporal features of financial data [39]. RL can be modeled through the Markov decision process, and the Q value function is calculated by the Bellman equation;

however, the calculation of the Q value function by the Bellman equation is not applicable when the state action space is large; therefore a deep neural network is more suitable for use in approximating Q value functions. In the MSF-DRL framework, different neural networks capture the time-series dependence of multi-type data, and realize the RL learning process based on the neural network, which provides a theoretical basis for the realization of Algorithm 1.

4 Experimental setup

This section mainly introduces the experimental setup of this paper, including the dataset, performance evaluation metrics, comparison methods and implementation details of the proposed approach.

4.1 Datasets

We validate our trading strategy using historical data from stocks jointly composed of the China and the S&P 500 stock market indexes. The experimental dataset is obtained from Yahoo Finance². We are comparing eight trading strategies, when comparing with the first seven trading strategies, the training period ranges from January 1, 2010, to January 1, 2018, and the test period ranges from January 1, 2018, to January 1, 2021. We compare another trading strategy [17] only on the S&P 500 index, and the training period and test period are consistent with the settings in [17]. For all datasets, the stock data structure can be found in Table 2, mainly including the opening price, highest price, lowest price, adjusted closing price and trading volume.

4.2 Metrics

We compare our trading strategy based on MSF-DRL framework with the other trading strategies through some evaluation metrics. The experimental evaluation metrics in this paper include profit curves, total profits (TP), AR, SR, MDD and max drawdown duration (MDDD). The details are as follows.

1. The total profits (TP) represents the return obtained after a period of trading, which is an essential quantitative metric measuring stock trading.
2. The annualized return (AR) refers to the ratio of the income obtained during a one-year investment period to

²<https://finance.yahoo.com/>

Table 2 Datasets structure example

Date	Open	High	Low	Close	Volume
...	...	...	...	...	...
2016/4/5	151.43	151.60	149.65	120.28	3337000
2016/4/6	149.90	150.15	147.81	120.30	4194600
2016/4/7	149.07	149.60	148.00	118.88	3663000
2016/4/8	149.06	149.99	148.12	119.76	3293800
...	...	...	...	...	...

the principal, and the corresponding calculation formula is as follows:

$$AR = \frac{\text{Total Profits}}{\text{Principal}} * \frac{365}{\text{Trading Days}} * 100 \quad (15)$$

3. The Sharpe ratio (SR) is a fund performance evaluation standardization index that is a measure of the profits exceeding the risk-free rate for each unit of risk taken. The higher the SR value is, the higher the profit obtained under certain risk conditions. In contrast, if the SR value is small or even negative, the profits obtained by taking a certain risk are small or zero. The corresponding calculation formula is as follows:

$$SR = \frac{E(R_P) - R_f}{\sigma_P} \quad (16)$$

where $E(R_P)$ is the expected portfolio return, R_f is the risk-free rate, and σ_P is the standard deviation of the return.

4. The max drawdown (MDD) refers to the maximum value of the return rate retracement range when the net value of the product reaches the lowest point at any historical point in the selected period, this is an important risk indicator, where the larger the value, the greater the risk.
5. The max drawdown duration (MDDD) describes the time from the beginning of the drawdown to the new high of the holding value.

4.3 Baselines

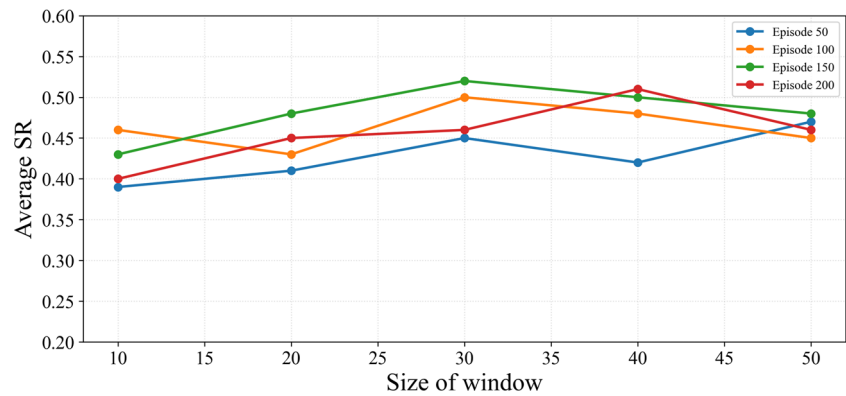
In this section, we compare our MSF-DRL framework with other algorithmic trading methods on real transaction data.

1. Buy and Hold (B&H) is a traditional trading strategy in which after a certain investment portfolio is constructed according to a determined and appropriate asset allocation ratio, the asset allocation status is not be changed during the holding period.
2. The two RL models for stock trading based on Q-learning in [11] use the clustering method and candlestick to represent the state of the stock market. As mentioned in the article, in most cases, the performance of model 1 is better than that of model 2, so in this article, we use only model 1 for comparison. In model 1, the size n of the clusters is set to 3, 6, and 9, and we compare them.
3. CNN-TA [13] is a deep learning model, which uses 15 different technical indicators and converts them into 15×15 2D images through CNN analysis of 2D images to determine the buying and selling point to implement algorithmic trading.
4. Fuzzy deep direct reinforcement learning (FDDR) [14] is a DRL model that introduces fuzzy learning to reduce the uncertainty of input data. It is the first paper to implement DL in the design of financial signal representation and self-taught RL.
5. Trading deep Q-network (TDQN) [15] is a DRL model that treats algorithmic trading as an RL process to learn dynamic trading strategies.
6. The time-driven feature-aware jointly deep reinforcement learning (TFJ-DRL) [16] model, which used GRU to extract temporal features of stock data and technical indicators and implements autonomous trading through the policy gradient algorithm.
7. Parallel multi-module deep reinforcement learning (PMML) [18] is a DRL model in which the agent the perceives environment through the features extracted by

Table 3 Detailed information about MSF-DRL framework

No.	Type of layer	Kernel size	Number of Filters	Stride	Activation function
1	Convolutional Layer	3×3	32	1	Relu
2	Max Pooling	2×2	—	2	—
3	Convolutional Layer	3×3	32	1	Relu
4	Max Pooling	2×2	—	2	—
5	Convolutional Layer	3×3	64	1	Relu
6	Max Pooling	2×2	—	2	—
7	BiLSTM	—	128	—	—
8	LSTM	—	128	—	—
9	FC	—	128	—	Leaky_relu

Fig. 5 Changes between window size and average SR



the parallel FC and LSTM modules and makes trading decisions.

8. HW_LSTM_RL [17] is a structure that combines wavelet transformation and DRL for stock trading, and is a relatively new method. Since Lee et al. [17] compared only the SR values of HW_LSTM_RL and its variant structure and B&H, the SR values of HW_LSTM_RL were all higher than that of its variant structure, so we compare only the SR values of HW_LSTM_RL and B&H in this paper.

4.4 Implementation details of the proposed approach

MSF-DRL framework is based on a combination of deep learning and RL. This experiment was performed using the PyTorch framework, and the specifications of the computer used in the experiment are as follows: CPU, Intel Core i9-9820X; GPU, NVIDIA GeForce RTX 2080 Ti; and memory, 64GB (DDR4). In the deep learning perception module, for stock data and technical indicators, a two-layer LSTM network is used directly to extract features, where the size of the LSTM hidden layer is 128. The image size is 393×287 , each image contains 30 days of stock price and trading

volume information, and features are successively extracted through a three-layer CNN and BiLSTM; a smaller size filter can capture more details of the image, 3×3 is the smallest and most commonly used kernel size in image processing applications [13], so a 3×3 filter size is used in this paper; some related parameter settings are shown in Table 3. There are 2,013 images in the training dataset and 756 images in the test dataset for each stock.

In the process of learning the optimal trading strategy, we set the number of episodes to 50, 100, 150 and 200 respectively, and conducted comparative experiments based on different windows sizes. It can be seen from Fig. 5 that for stocks with different price trends, when the window size is 30 and the episode is 150 the highest average SR can be obtained, so the window size is set to 30, the number of episodes is set to 150. Similarly, we get that relatively good trading results can be obtained when the size of the BiLSTM and LSTM hidden layers is 128, and the learning rate is 0.0001. In RL, Adam optimizer is used to improve both the training stability and the convergence speed of the MSF-DRL framework. In ϵ -greedy strategy, the initial ϵ is 0.01, and as the number of trading actions increases, the value of ϵ gradually increases to the maximum value of 1. To prevent overfitting, the normalization operation is used

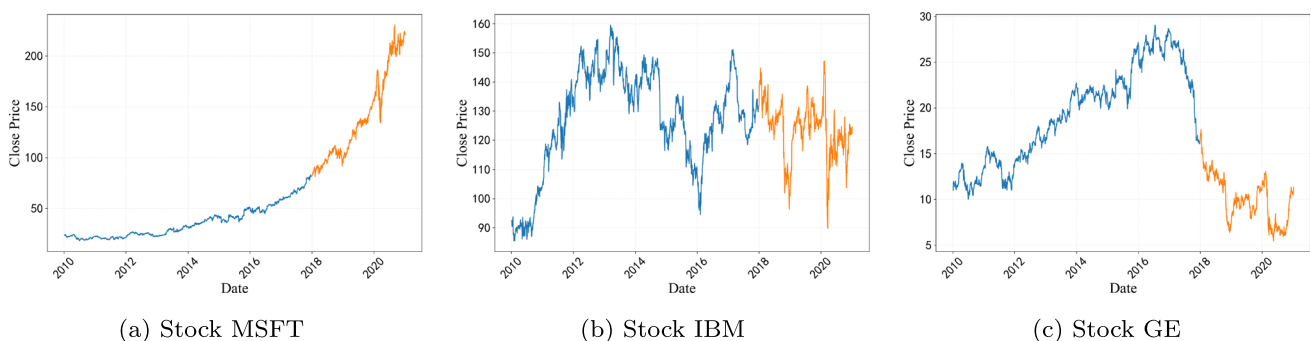


Fig. 6 Three stocks with different price trends (blue parts: model training, orange parts: out-of-sample tests. The stock price in the test stage of (a) shows an upward trend, the stock price in the testing stage of (b)

shows a trend of fluctuation, and the stock price in the testing stage of (c) shows a downward trend)

in the network, and dropout is used, the ratio of which is 0.2. In the transaction process, the transaction cost is an issue that cannot be ignored. Frequent buying and selling transactions generate excessive transaction costs. Taking this into consideration, we set the transaction cost to 0.1% as set in most studies.

5 Experiments

In this section, we first conduct experiments on the dataset containing three different stock price changes (rising, falling and no obvious direction), and the change in the price (closing price) of each stock in the dataset is shown in Fig. 6.

Fig. 7 Profit curves of three stocks with different price trends (Due to the fact that the return values of different trading strategies are very close, only part of the yield curve will appear in the yield curve chart)

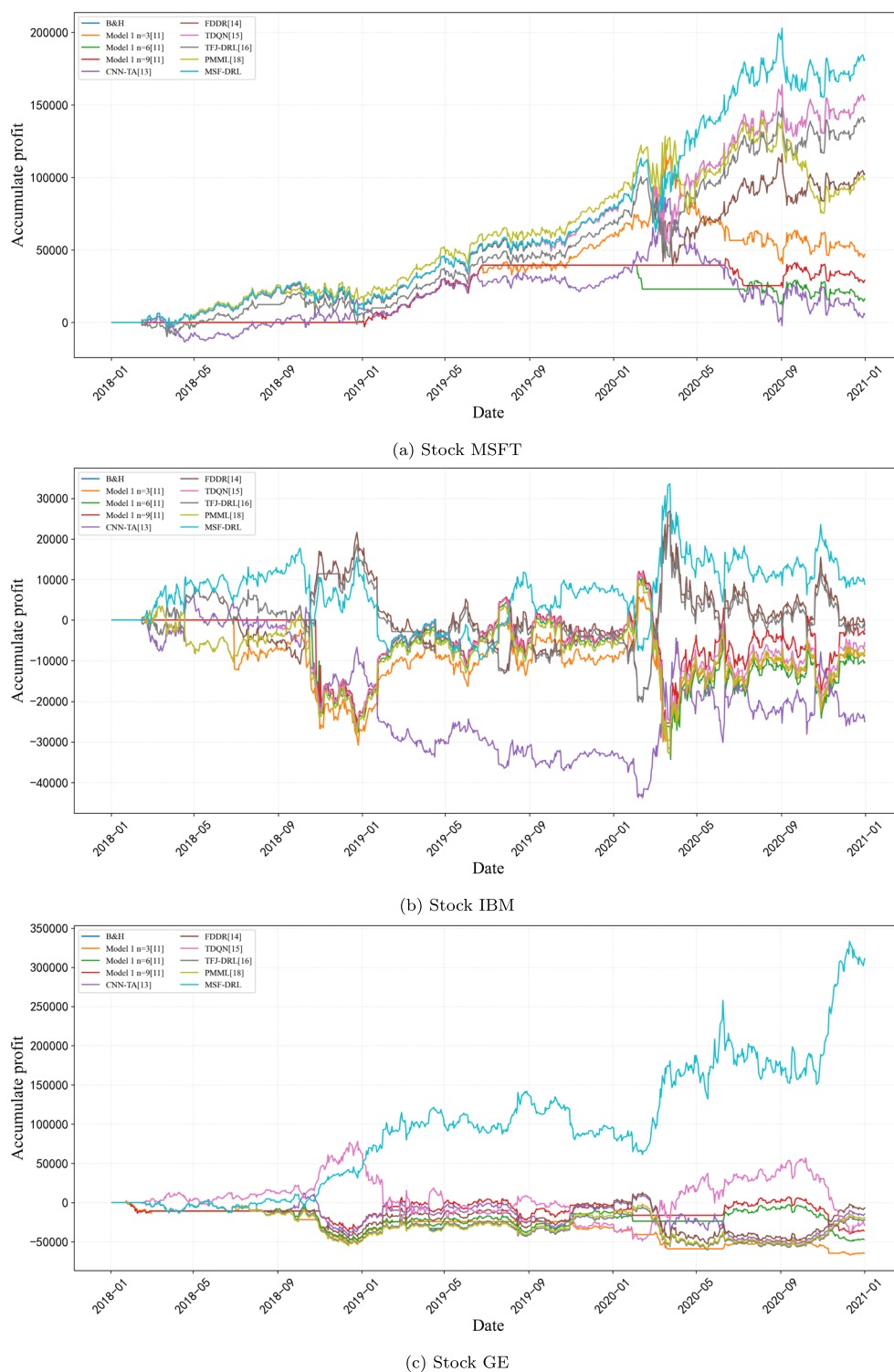


Table 4 Experimental results of three different stock price trends

Method	MSFT					IBM					GE				
	TP	AR (%)	SR	MDD (%)	MDDD	TP	AR (%)	SR	MDD (%)	MDDD	TP	AR (%)	SR	MDD (%)	MDDD
B&H	154105.00	27.75	1.14	52.78	24	-7269.84	1.81	0.06	433.41	26	-21889.13	3.39	0.07	1897.21	499
Model 1 n=3 [11]	46997.92	20.69	0.64	53.81	202	-8744.09	1.89	0.04	33.76	20	-64675.04	-65.81	-0.95	67.58	726
Model 1 n=6 [11]	16268.97	8.67	0.39	28.34	386	-10393.7	1.32	0.03	40.72	27	-47101.25	-25.49	-0.44	49.84	237
Model 1 n=9 [11]	29052.92	14.02	0.70	30.60	141	-3275.79	4.55	0.11	33.79	20	-36120.83	-15.22	-0.29	43.97	62
CNN-TA [13]	5618.55	7.11	0.33	47.51	119	-24957.33	-4.93	-0.15	46.87	449	-14559.55	6.04	0.13	58.13	485
FDDR [14]	102501.22	23.11	0.88	34.79	33	-1332.27	3.16	0.11	34.39	285	-6907.55	8.59	0.19	66.67	65
TDQN [15]	98983.78	22.33	0.90	27.09	64	-7268.23	1.80	0.06	38.95	27	-19052.55	4.48	0.09	60.34	499
TFJ-DRL [16]	153950.07	27.73	1.14	28.03	24	-5564.99	1.93	0.05	36.27	93	-28995.11	7.46	0.13	71.36	285
PMML [18]	139002.58	26.48	1.07	30.04	35	-2647.71	2.36	0.09	32.65	133	-21851.16	3.40	0.07	61.72	249
MSF-DRL	181489.00	29.65	1.27	44.92	14	8878.35	5.34	0.24	86.68	53	311396.17	41.34	1.35	41.56	96

Bold entries signify the best results

To prove the generalization ability of the MSF-DRL framework on the stock market, we further conduct experiments on a dataset containing more stocks. In addition, we compare with the HW_LSTM_RL method. Moreover, we compare with the other two fusion strategies on a dataset consisting of three stocks with different trends, and finally conduct ablation experiments to test the validity of the MSF-DRL framework.

5.1 Experimental comparison of three different trends in stocks

For three stocks with different price trends, we compare our trading strategy with the other trading strategies. We compare the yield curve and evaluation index results, as shown in Fig. 7 and Table 4, respectively.

In regard to the traditional trading strategy B&H is a passive trading strategy that has advantages for stocks with rising prices. However, the trading strategy of B&H does not make judgments about the stock market; thus, it does not obtain good returns on stocks with relatively large price fluctuations. From the experimental results, we can see that for MSFT, whose price is on the rise, B&H can obtain better profits. Nevertheless, for IBM and GE, whose stock prices fluctuate normally and fall, this trading strategy does not perform better than the other trading strategies. The trading strategies learned based on model 1, which can select the

trading behavior through self-learning, are more flexible regarding the choices of trading behavior for stocks with fluctuating stock prices than the traditional trading strategy of B&H. However, the trading strategy based on model 1 lacks the ability to perceive the stock market environment, which has an impact on the correct choice of trading behavior and affects the final profits. For stock MSFT whose prices are rising, the trading strategy learned by the CNN-TA model can yield benefits overall, compared with those from other trading strategies, but there is no obvious advantage. Compared with the trading strategies learned based on RL or deep learning, the trading strategies learned by the DRL model perform better for three different stocks with different trends. In the DRL model, for stocks with different trends, compared with the trading strategies learned by FDDR and TDQN, which do not address the temporal relationship of stock data, the trading strategies learned by TFJ-DRL, PMML and MSF-DRL that address the temporal relationship of stock data are more advantageous. For stock MSFT, the SR of TFJ-DRL, PMML and MSF-DRL all exceed 1, indicating that the trading strategies can obtain more returns under the same investment risk. Moreover, compared with TFJ-DRL and PMML, MSF-DRL perceives the stock market environment by analysing multi-type data (stock data, technical indicators and candlestick charts), stock data are often used to analyse the stock market, but there is noise in stock data, the combination of stock data and technical indicators can reduce the impact of noise in

Table 5 Larger datasets of stocks in different industries

Technology	Financial	Energy	Healthcare	Industrials	Automotive
BIDU	000001	601088	600276	600019	600104
BABA	601939	000983	300015	000708	000625
0700.HK	601288	PTR	ABC	EMR	000996
GOOGLE	BAC	TOT	JNJ	EXPD	TSLA
NVDA	JPM	XOM	MRK	MA	GM

Table 6 TP results comparison (The stocks we selected are traded in CNY and USD, so we calculated average returns for the two currencies respectively, the first represents average return of CNY, the second represents average return of USD)

Sectors	Stock	B&H	model 1(3) [11]	model 1(6) [11]	model 1(9) [11]	CNN-TA [13]	FDDR [14]	TDQN [15]	TFJ-DRL [16]	PMML [18]	MSF-DRL
Technology	BIDU	5000	11846.45	-5739.79	4972.05	3871.05	13820.16	8411.73	14692.71	15390.27	21402
	BABA	24448	40704.55	76865.66	41609.00	68241.43	81038.23	8818.00	78146.91	83720.45	89261
	0700.HK	34845	27823.71	73139.96	25806.47	42072.16	69560.42	79650.42	73902.61	76332.84	81885
GOOGLE	63135	30747.41	45350.09	72337.25	76409.82	76340.05	81006.19	85731.42	88461.27	101874	101874
	NVDA	117787	12195.00	32456.00	-54661.78	-31843.23	144291.35	-70950.56	184732.52	197241.58	243562
Financial	000001	70270	52253.24	26853.04	14238.83	66341.39	75414.30	72175.97	73521.03	71392.41	79615
	601939	-13703	-4198.17	-2681.23	7804.49	-3278.93	-4429.02	-8550.17	-3279.04	-2485.15	9062
	601288	-15189	-10256.40	-7839.21	-18882.47	-16399.24	-17897.36	-15188.25	-11083.20	-9735.00	-1360
BAC	1220	13413.70	11535.76	11427.48	13981.32	35142.63	40339.57	37242.06	41642.63	58747	58747
	JPM	20015	-26531.20	-21276.58	-22713.41	9825.48	16352.28	10127.72	18632.58	21405.43	28495
	601088	-10554	-8700.49	-8656.06	-22200.00	-11672.33	-6783.45	-10554.39	-2091.41	3481.73	27868
Energy	000983	-13369	-46564.83	-37181.00	-41555.35	-27968.92	-12730.52	-13368.89	-8936.70	-10424.31	5555
	PTR	-49034	-35431.45	-19339.34	-21475.24	-48749.32	-13872.16	-46229.38	-11394.72	-9358.31	48835
	TOT	-12911	-11750.66	-8015.87	-12503.09	-11721.05	-7413.25	-6022.23	-7290.23	-5981.43	18262
XOM	-37154	-13037.90	-26950.24	-22276.93	-14903.71	-9241.48	-7604.05	-6592.21	-5928.20	-1528	-1528
	600276	108452	39081.52	41927.59	108648.38	86451.23	97473.08	108434.19	98625.03	91470.51	106029
	300015	470075	33266.00	22003.00	-20102.58	301301.26	387013.55	339278.13	368102.34	410530.24	509988
ABC	6922	20472.92	23206.51	12055.00	13055.38	20834.39	-26030.11	22573.10	23730.19	25746	25746
	JNJ	31542	20627.11	8275.42	8776.48	13879.13	27401.31	-20099.90	30491.00	32637.41	27354
	MRK	62585	9939.36	9460.06	9939.35	37241.87	62041.58	63178.03	63024.20	68243.14	65099
Industrials	600019	-28470	-35950.43	-35629.52	-32087.30	-14291.03	-39671.54	-31887.97	-32574.51	-36120.37	1210
	000708	290808	4613.64	-32896.83	66894.90	80167.28	252849.31	237213.70	295914.30	302905.43	371863
	EMR	22862	34309.29	19005.03	20558.36	29701.63	32823.57	31535.54	30781.57	34056.32	39853
EXPD	55050	55461.46	20411.23	33177.48	32136.05	39202.30	37104.06	38413.02	40282.37	39590	39590
	MA	109687	-29965.32	45047.43	11928.18	69836.62	90350.92	109613.03	93821.62	95309.30	119785
	600104	-11867	-34440.34	-32752.56	-33349.16	-25237.18	-8248.15	-11864.29	-7492.38	-6838.21	12502
Automotive	000625	93882	-66802.66	-61894.78	-62632.20	57810.72	85215.32	93878.06	96303.41	90741.19	106411
	000996	-38094	23344.83	33554.95	34497.45	-27441.80	-19163.02	-39173.99	18362.08	14942.01	37894
	TSLA	993131	174173.03	80638.27	952837.67	52913.04	183941.63	71611.58	874280.01	973149.83	1497936
GM	9235	17809.69	2949.45	-60823.96	8562.23	12483.74	13525.52	14071.26	16308.53	22631	22631
Average											
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stock data, the candlestick chart generated by the K-line theory standardizes the data, reduce the impact of noise, and can provide useful trading signals. The three reflect the stock market information from different perspectives, in the MSF-DRL framework, different neural networks are used to extract their temporal features, and the agent in RL analyses the stock market according to the fused features, learns relatively better trading strategies and obtains higher TP, AR and SR.

Compared with the traditional trading strategy B&H and other trading strategies learned based on deep learning, RL

or DRL, which analyze the stock market based on stock data or technical indicators, we combine deep learning and RL and integrate the features of stock data, technical indicators and candlestick charts through the MSF-DRL framework to obtain a deeper expression of the environment of the stock market and to learn the optimal trading strategy. The experimental results confirm that our trading strategy can profit from stocks with different price trends and performs better than the other trading strategies; for stock MSFT whose prices are increasing, all trading strategies obtain good returns, while for the IBM stock with fluctuating prices,

Table 7 AR results comparison

Sectors	Stock	B&H	model 1(3) [11]	model 1(6) [11]	model 1(9) [11]	CNN-TA [13]	FDDR [14]	TDQN [15]	TFJ-DRL [16]	PMML [18]	MSF-DRL
Technology	BIDU	4.58	12.24	6.05	10.01	5.14	4.58	9.63	10.01	9.87	12.46
	BABA	12.00	20.07	24.12	20.02	20.21	21.73	20.48	19.23	22.30	25.14
	0700.HK	13.37	14.49	22.38	14.99	15.08	15.72	17.20	17.46	17.52	22.81
	GOOGLE	17.58	17.05	19.11	22.97	19.83	21.72	20.02	20.41	21.30	24.48
	NVDA	28.88	16.90	19.10	-28.52	-17.35	24.62	-12.71	25.12	26.32	37.51
Financial	000001	18.89	19.76	14.82	10.56	17.25	19.33	19.15	17.41	18.03	20.12
	601939	-2.65	1.57	1.82	5.01	0.93	0.61	-0.65	1.67	2.03	5.02
	601288	-4.33	-3.54	-2.27	-7.88	-8.34	-6.91	-4.33	-3.45	-4.69	0.87
	BAC	7.30	14.94	13.17	14.05	14.69	14.85	15.86	15.28	16.01	17.66
	JPM	11.00	-10.23	-6.97	-7.95	6.04	9.33	8.50	10.17	10.49	12.67
Energy	601088	-0.46	-1.15	0.51	-7.96	-1.82	-1.30	-0.46	0.37	1.24	9.90
	000983	0.34	-6.48	-9.89	-6.44	-2.61	-3.12	-0.33	0.95	-1.47	6.49
	PTR	-21.18	-15.09	-4.54	-5.61	-20.73	-3.82	-19.24	-4.29	-3.60	14.00
	TOT	2.12	2.07	2.99	0.88	0.71	3.38	4.56	3.95	5.40	8.64
	XOM	-10.74	1.21	-5.76	-2.69	-4.25	0.82	1.30	1.39	0.95	1.56
Healthcare	600276	24.57	19.71	20.14	33.82	22.90	23.81	24.57	24.32	21.30	24.32
	300015	43.59	18.82	13.17	-2.82	35.59	36.71	32.86	34.62	39.46	44.60
	ABC	7.07	15.16	14.76	11.99	10.73	13.31	-5.04	12.07	14.21	15.39
	JNJ	10.61	11.36	7.33	7.81	8.30	7.55	-5.01	10.16	12.37	9.71
	MRK	16.22	7.33	7.17	7.33	14.81	16.48	17.29	16.83	18.04	16.60
Industrials	600019	-7.97	-14.30	-14.26	-12.17	9.94	-16.73	-14.64	-13.25	-15.41	4.00
	000708	36.84	46.40	3.55	32.91	31.04	38.01	34.17	37.31	39.72	40.09
	EMR	11.95	20.18	15.62	14.91	17.72	18.64	14.65	16.38	18.20	21.06
	EXPD	15.49	23.34	12.74	17.16	13.58	12.74	12.33	14.92	14.37	12.79
	MA	24.27	-15.84	20.31	10.27	19.14	21.75	24.26	20.25	22.30	24.85
Automotive	600104	1.04	-15.93	-14.76	-15.85	-12.52	1.50	0.23	2.46	4.69	8.43
	000625	25.40	-71.84	-50.49	-55.16	19.82	21.91	25.40	23.28	20.89	26.72
	000996	-4.17	16.53	21.59	19.23	-3.94	-4.15	-4.79	10.81	6.37	22.77
	TSLA	59.86	55.49	40.43	85.81	31.21	53.82	40.85	59.32	55.23	64.91
	GM	10.50	12.53	10.81	-52.01	8.63	11.28	10.26	9.72	12.70	13.57
Average		11.73	7.09	6.76	4.48	8.84	13.97	10.04	13.83	14.21	18.97

Bold entries signify the best results

only our trading strategy gains profits, while other trading strategies have different degrees of loss, especially when the stock GE price drops, our trading strategy can still profit, while the other strategies result in a loss. Additionally, our trading strategy has higher values in terms of TP, AR and SR than the other trading strategies. For three stocks with different price trends, our SR values reached 1.27, 0.24 and 1.35, and these values are significantly higher than those of the other trading strategies. The trading actions of our trading strategy can be long and short, and sometimes the indicators MDD and MDDD of our modal are not the best of the all trading strategies. In addition, it can be seen from the experimental results in Table 4 that the trading strategy learned

based on the DRL model is better than the trading strategy learned based on a single deep learning or RL model.

In other words, the MSF-DRL framework is evaluated through experiments on practical trading data and can obtain considerable profits for stocks with different price trends.

5.2 Experimental comparison of relatively large datasets

To further verify the generalization ability of the MSF-DRL framework, we select stocks from different industries for further training and testing. The specific industries and their corresponding stocks are shown in Table 5. We

Table 8 SR results comparison

Sectors	Stock	B&H	model 1(3) [11]	model 1(6) [11]	model 1(9) [11]	CNN-TA [13]	FDDR [14]	TDQN [15]	TFJ-DRL [16]	PMML [18]	MSF-DRL
Technology	BIDU	0.12	0.28	0.12	0.21	0.10	0.23	0.12	0.26	0.26	0.36
	BABA	0.38	0.52	0.78	0.53	0.58	0.64	0.26	0.59	0.72	0.81
	0700.HK	0.47	0.47	0.84	0.46	0.51	0.74	0.62	0.85	0.86	0.88
	GOOGLE	0.69	0.45	0.71	0.84	0.62	0.79	0.81	0.87	0.85	0.93
	NVDA	0.78	0.31	0.44	−0.40	−0.21	0.86	0.73	0.91	0.92	1.09
Financial	000001	0.71	0.73	0.43	0.30	0.62	0.71	0.72	0.73	0.69	0.76
	601939	−0.12	0.05	0.06	0.24	0.04	0.02	−0.03	0.04	0.06	0.25
	601288	−0.25	−0.17	−0.11	−0.31	−0.29	−0.27	−0.25	−0.13	−0.10	0.05
	BAC	0.20	0.29	0.27	0.28	0.26	0.39	0.49	0.42	0.51	0.63
	JPM	0.35	−0.24	−0.18	−0.20	−0.23	0.29	0.27	0.33	0.36	0.41
Energy	601088	−0.02	−0.04	0.01	−0.23	−0.06	−0.01	−0.02	0.03	0.11	0.46
	000983	0.01	−0.81	−0.17	−0.08	−0.13	0.01	0.01	0.05	0.02	0.21
	PTR	−0.51	−0.33	−0.11	−0.13	−0.39	−0.17	−0.50	−0.15	−0.07	0.68
	TOT	0.06	0.04	0.06	0.02	0.03	0.11	0.13	0.10	0.19	0.34
	XOM	−0.28	0.04	−0.11	−0.05	0.03	0.07	0.09	0.08	0.06	0.08
Healthcare	600276	0.84	0.53	0.55	0.98	0.78	0.80	0.84	0.82	0.79	0.83
	300015	1.75	0.47	0.39	−0.06	1.46	1.63	1.38	1.51	1.67	1.83
	ABC	0.23	0.35	0.38	0.28	0.29	0.34	−0.15	0.35	0.34	0.39
	JNJ	0.51	0.41	0.23	0.24	0.35	0.48	−0.21	0.53	0.43	0.47
	MRK	0.81	0.25	0.25	0.25	0.58	0.82	0.81	0.80	0.84	0.83
Industrials	600019	−0.26	−0.30	−0.29	−0.27	−0.17	−0.31	−0.32	−0.27	−0.28	0.15
	000708	1.39	0.46	0.04	1.37	1.34	1.29	1.27	1.34	1.39	1.57
	EMR	0.37	0.45	0.34	0.35	0.36	0.42	0.40	0.39	0.45	0.47
	EXPD	0.69	0.69	0.37	0.49	0.42	0.59	0.53	0.56	0.61	0.56
	MA	0.88	−0.52	0.61	0.27	0.69	0.83	0.88	0.83	0.85	0.91
Automotive	600104	0.03	−0.39	−0.37	−0.42	0.01	0.08	0.03	0.09	0.11	0.28
	000625	0.69	−0.95	−0.76	−0.85	0.53	0.61	0.69	0.70	0.65	0.74
	000996	−0.08	0.38	0.45	0.47	−0.03	−0.05	−0.09	0.14	0.09	0.49
	TSLA	1.51	0.85	0.63	1.71	0.39	0.82	1.26	1.52	1.73	1.75
	GM	0.28	0.31	0.21	−0.81	0.24	0.27	0.50	0.31	0.40	0.37
Average		0.41	0.15	0.20	0.18	0.28	0.45	0.37	0.49	0.52	0.65

Bold entries signify the best results

again compare our trading strategy with the other trading strategies and evaluate them with five indicators. The experimental results are shown in Tables 6, 7, 8, 9 and 10.

The 30 stocks in the experiment are selected from six different fields, including technology, financial, energy, healthcare, industrials and automotive. Each field is composed of stocks from the China and S&P 500 stock market indexes. As seen from the experimental results, in terms of overall performance, our trading strategy has a great advantage. From the average of the five evaluation

metrics, we can see that of the trading strategies tested, our trading strategy is more advantageous, and the average SR value of our trading strategy is 0.65, significantly higher than others. In particular, when stock NVDA is traded, our profit is the highest of the all trading strategies, with an SR value of 1.09, the highest of all the trading strategies.

Overall, for stocks with different trends, the performance of our trading strategy differs, and ultimately, some profits can be obtained. Even if there is a loss, the loss of our trading strategy is lower than those of the other trading strate-

Table 9 MDD results comparison

Sectors	Stock	B&H	model 1(3) [11]	model 1(6) [11]	model 1(9) [11]	CNN-TA [13]	FDDR [14]	TDQN [15]	TFJ-DRL [16]	PMML [18]	MSF-DRL
Technology	BIDU	418.09	53.04	43.20	50.92	47.29	54.61	64.49	50.37	52.53	35.01
	BABA	73.08	39.95	44.65	40.90	51.72	43.63	46.67	37.86	40.36	29.99
	0700.HK	471.54	29.04	44.95	39.57	43.19	36.13	46.73	34.92	28.48	23.15
	GOOGLE	104.35	36.51	44.89	51.84	49.37	38.49	23.17	32.34	26.30	19.21
	NVDA	336.94	57.69	61.03	68.09	62.89	79.46	84.36	69.37	59.81	56.01
Financial	000001	89.41	42.59	31.66	38.48	29.22	32.61	29.44	34.72	30.18	30.76
	601939	512.73	21.74	17.62	17.62	19.47	22.85	25.10	20.32	18.27	16.72
	601288	364.91	16.54	16.54	19.41	18.61	19.29	20.68	17.39	16.65	15.74
	BAC	356.75	48.92	47.01	48.92	50.39	52.64	55.73	47.28	51.39	19.50
	JPM	196.30	43.41	37.16	37.15	40.72	37.95	36.58	39.27	35.51	35.76
Energy	601088	507.67	24.07	31.68	34.02	43.89	28.86	44.77	30.29	28.06	19.95
	000983	980.00	53.98	42.49	48.73	44.78	48.35	49.09	46.84	49.05	41.03
	PTR	353.64	44.03	34.38	38.94	52.86	41.72	55.78	35.96	31.46	25.04
	TOT	381.35	44.53	43.54	51.25	50.38	47.14	50.45	44.21	48.80	36.36
	XOM	455.26	41.17	48.02	44.69	42.83	39.62	48.88	41.20	38.59	33.15
Healthcare	600276	152.51	45.57	46.00	53.39	48.64	47.01	46.50	46.37	48.92	45.50
	300015	43.43	41.10	37.55	58.73	32.92	30.84	28.38	26.50	24.49	12.21
	ABC	645.09	23.78	23.77	28.95	29.78	21.83	38.59	26.92	29.17	28.72
	JNJ	135.60	27.35	27.34	27.33	29.34	28.75	52.48	38.24	30.46	29.29
	MRK	62.55	27.27	27.27	27.27	56.82	29.43	67.05	27.48	25.60	26.87
Industrials	600019	871.93	42.78	41.07	44.28	45.65	43.72	53.25	41.82	39.26	47.14
	000708	29.51	61.39	53.17	50.14	55.37	33.86	21.38	28.56	25.31	21.38
	EMR	377.56	42.42	49.70	40.27	39.69	42.48	36.87	45.37	38.09	41.63
	EXPD	137.31	36.33	32.45	32.00	35.64	31.79	32.44	30.53	31.82	27.33
	MA	81.32	45.69	33.05	36.00	37.92	34.21	40.97	36.74	32.41	31.98
Automotive	600104	355.82	40.55	41.35	41.87	43.28	35.59	47.35	32.12	29.68	26.44
	000625	39.75	66.80	62.05	62.63	42.74	32.26	23.60	30.82	28.76	23.61
	000996	231.91	43.72	38.08	34.63	56.37	59.31	60.36	44.74	38.90	32.82
	TSLA	38.75	63.58	68.67	90.51	62.24	40.52	61.39	42.59	54.61	33.73
	GM	708.02	46.83	51.40	68.71	60.33	57.43	55.23	54.33	50.27	45.53
Average		317.10	41.75	40.72	44.24	43.61	39.87	44.89	37.85	35.00	30.39

Bold entries signify the best results

Table 10 MDDD results comparison

Sectors	Stock	B&H	model 1(3) [11]	model 1(6) [11]	model 1(9) [11]	CNN-TA [13]	FDDR [14]	TDQN [15]	TFJ-DRL [16]	PMML [18]	MSF-DRL
Technology	BIDU	462	226	225	226	243	239	262	223	217	199
	BABA	42	197	277	197	83	62	65	52	61	41
	0700.HK	153	196	125	131	167	152	139	146	138	112
	GOOGLE	23	69	69	69	36	30	23	28	26	14
	NVDA	58	151	202	265	127	142	162	115	103	55
Financial	000001	115	35	56	107	65	68	115	60	72	95
	601939	91	115	91	91	106	84	91	87	91	79
	601288	541	360	360	537	381	355	541	340	347	325
	BAC	55	55	55	55	62	49	69	50	55	22
	JPM	55	251	48	48	53	47	62	50	52	46
Energy	601088	538	560	535	535	528	421	538	350	229	80
	000983	566	241	241	360	319	339	566	317	285	235
	PTR	367	554	554	554	463	524	490	342	298	144
	TOT	368	555	554	251	357	318	364	279	260	181
	XOM	375	422	557	557	417	396	365	380	391	379
Healthcare	600276	149	35	35	30	132	127	149	116	105	149
	300015	82	57	101	446	72	64	89	59	43	12
	ABC	198	23	22	245	148	162	137	142	119	121
	JNJ	32	32	32	32	33	37	35	33	29	31
	MRK	55	55	55	55	57	62	71	60	65	53
Industrials	600019	540	537	688	537	518	523	540	482	501	336
	000708	69	122	62	134	83	76	69	62	79	58
	EMR	45	45	45	38	48	47	68	45	39	35
	EXPD	44	8	44	80	65	59	44	40	48	44
	MA	23	687	90	534	31	29	23	27	25	21
Automotive	600104	525	686	690	690	517	523	525	501	492	243
	000625	20	686	686	695	22	26	20	24	20	17
	000996	267	187	265	187	148	153	267	164	142	132
	TSLA	5	8	80	5	43	58	116	37	26	30
	GM	445	136	443	339	248	139	85	128	138	124
Average		210.26	243.03	242.90	274.33	185.73	183.26	202.33	161.47	149.87	113.77

Bold entries signify the best results

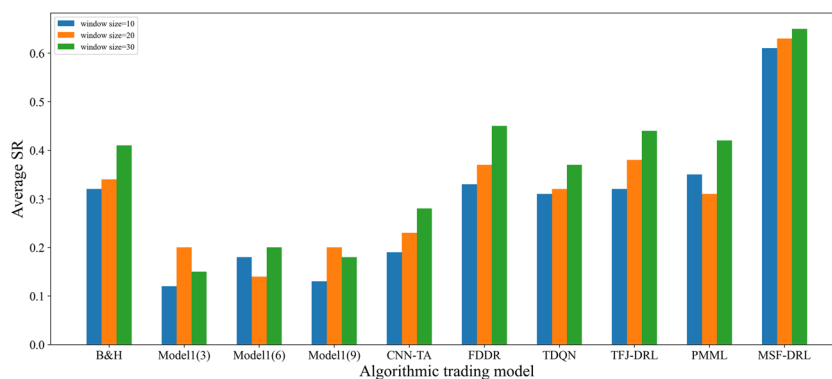
Fig. 8 Average SR change of different methods under different window sizes

Table 11 Var, Max, Min and Average of SR with different methods under different time window sizes (w represents the time window size)

Method	Var			Max			Min			Average		
	w=10	w=20	w=30	w=10	w=20	w=30	w=10	w=20	w=30	w=10	w=20	w=30
B&H	0.25	0.27	0.28	1.47	1.46	1.75	−0.56	−0.63	−0.51	0.32	0.34	0.41
Model 1 n=3 [11]	0.24	0.21	0.19	0.81	1.12	0.85	−0.73	−0.62	−0.95	0.12	0.20	0.15
Model 1 n=6 [11]	0.25	0.22	0.13	1.02	1.24	0.84	−0.70	−0.82	−0.76	0.18	0.14	0.20
Model 1 n=9 [11]	0.27	0.28	0.30	1.22	1.38	1.71	−0.69	−0.43	−0.85	0.13	0.20	0.18
CNN-TA [13]	0.22	0.19	0.17	1.05	1.14	1.46	−0.57	−0.33	−0.39	0.19	0.23	0.28
FDDR [14]	0.22	0.20	0.21	1.53	1.69	1.63	−0.64	−0.42	−0.31	0.33	0.37	0.45
TDQN [15]	0.23	0.26	0.24	1.36	1.41	1.38	−0.43	−0.59	−0.50	0.31	0.32	0.37
TFJ-DRL [16]	0.21	0.22	0.20	1.14	1.29	1.52	−0.40	−0.37	−0.27	0.32	0.38	0.44
PMML [18]	0.24	0.22	0.21	1.62	1.50	1.73	−0.33	−0.26	−0.28	0.35	0.31	0.42
MSF-DRL	0.21	0.20	0.19	1.69	1.75	1.83	0.01	−0.03	0.05	0.61	0.63	0.65

Bold entries signify the best results

gies. For the overall average TP, AR, SR, MDD and MDDD, it can be further illustrated that the trading strategies learned based on the DRL model have more advantages, and the DRL models that focus on temporal relationships are better than other models. Moreover, our trading strategy achieves the best results of the trading strategies, which indicates that our trading strategy has better performance generalization ability and profitability. In addition, the SR of our trading strategy is the highest, which means that our trading strategy can achieve higher returns under the same level of risk.

Further, we set the window size of each method to 10, 20 and 30 and verify the sensitivity of different methods to data by comparing the changes in the average SR value. From the experimental results shown in Fig. 8, it can be seen that the difference in window size will cause the final result to fluctuate. Compared with the trading strategies learned by other models, the trading strategies learned under the MSF-DRL framework have the least fluctuations, indicating

Table 12 The results of the Mann-Whitney U test ($P1$ represents the results with time window sizes 10 and 20, $P2$ represents the results with window sizes 10 and 30, $P3$ represents the results with window sizes 20 and 30)

Method	$P1$	$P2$	$P3$
B&H	0.80	0.72	0.57
Model 1 n=3 [11]	0.04	0.13	0.43
Model 1 n=6 [11]	0.27	0.01	0.03
Model 1 n=9 [11]	0.02	0.21	0.35
CNN-TA [13]	0.33	0.03	0.04
FDDR [14]	0.47	0.02	0.03
TDQN [15]	0.03	0.46	0.23
TFJ-DRL [16]	0.12	0.04	0.25
PMML [18]	0.03	0.28	0.34
MSF-DRL	0.52	0.72	0.68

Table 13 SR value result comparison

Year	HW_LSTM_RL	B&H	MSF-DRL
Year 1	0.4962	0.3059	0.4985
Year 2	−0.4228	−0.9450	0.3606
Year 3	1.3129	0.7991	1.6854
Year 4	0.8987	0.6715	0.9780
Year 5	0.1829	0.0677	0.2413
Year 6	0.8345	0.9395	0.7364
Year 7	2.2331	2.2287	2.0638
Year 8	1.1665	1.0894	1.3243
Year 9	0.2283	0.0324	0.4852
Year 10	1.4318	0.8860	1.6764
Year 11	2.5242	2.5897	2.6105
Year 12	−0.2941	−0.3433	0.0025
Average	0.8827	0.6935	1.0552

Bold entries signify the best results

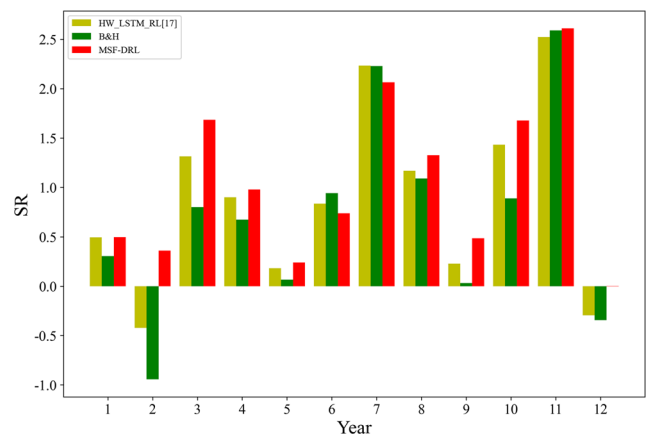
**Fig. 9** Histogram comparison of SR values (The SR value is too close to 0 at Year 12 to be shown in the figure)

Table 14 Experimental results of three different fusion strategies

Fusion strategies	MSFT					IBM					GE				
	TP	AR (%)	SR	MDD (%)	MDDD	TP	AR (%)	SR	MDD (%)	MDDD	TP	AR (%)	SR	MDD (%)	MDDD
Concatenation	158167.04	28.07	1.15	38.03	24	−34004.58	−10.98	−0.34	428.53	342	−69299.27	−42.79	−0.55	598.13	219
Bilinear Pooling	74134.35	19.53	0.74	54.62	24	−11505.61	0.28	0.01	847.67	27	21651.48	8.72	0.39	90.56	293
Add	181489.00	29.65	1.27	44.92	14	8878.35	5.34	0.24	86.68	53	311396.17	41.34	1.35	41.56	96

Bold entries signify the best results

that the data sensitivity of the MSF-DRL framework is relatively weak. Table 11 shows the var, min and max values of SR for different methods under different window sizes. It can be seen in the results that when the time window size is 10, 20, and 30, the var values of MSF-DRL are 0.21, 0.20 and 0.19, respectively, and compared with other methods, the values of var, min and max of SR under the MSF-DRL framework are relatively small changes. Moreover, for each method to obtain SR values at different time window sizes, we conduct Mann-Whitney U test, the results are shown in Table 12. In the Mann-Whitney U test, the significance level is set at 0.05, if P value is less than 0.05, the difference is considered to be significant, it can be seen from the Table 12 that under the different time window size, the SR values obtained by the MSF-DRL framework are not significant difference, it also shows that the MSF-DRL framework has relatively weak data sensitivity, although other method also has less sensitivity to the data, the SR values obtained by MSF-DRL framework are higher.

5.3 Comparison of the HW_LSTM_RL

In this section, we compare our MSF-DRL framework with a newer method. In this paper, we select only stocks of the S&P 500 index for comparison. According to the experiment in [17], two years is used for training and one year is used for testing, where Year 1 represents the training time from July 1, 2004, to December 31, 2006, and the test time is from January 1, 2007, to December

31, 2007. Year 2 represents the training time is from July 1, 2005, to December 31, 2007, and the test time from January 1, 2008, to December 31, 2008. In turn, 12 groups of comparative experiments are carried out, and the experimental comparison results are shown in Table 13. To more intuitively display the comparison results, we plot the experimental results as a histogram, as shown in Fig. 9.

In the structure HW_LSTM_RL, the wavelet transform is used to reduce the noise impact in the stock data, and the single stock data processed by the wavelet transform are analyzed in the implementation of algorithmic trading. Compared with the traditional trading strategy B&H, the trading strategy learned under the structure HW_LSTM_RL can obtain a higher SR value. In our MSF-DRL framework, we use relevant technical indicators to reduce the impact of stock data to capture the main trends of the stock market and to integrate the temporal features of the image data candlestick chart. Integration can obtain a deeper analysis of the stock market, which helps in the learning of the optimal trading strategy. The results demonstrate that our trading strategy can achieve the highest SR value in most cases, especially in Year 12, our SR value is greater than 0, while the other two trading strategies all have SR values less than 0, and our average SR value reaches 1.0552, which is significantly higher than those of the other two trading strategies.

5.4 Comparison of multi-type data fusion strategies

In this subsection, we compare with two other multi-type data fusion strategies, that is, concatenation and bilinear

Fig. 10 Profit curves of ablation experiments on stock GE



pooling. In MSF-DRL, we add the features of multi-type data to achieve fusion to verify the effectiveness of the multi-type data fusion method in MSF-DRL, we replace the fusion strategy in the MSF-DRL framework with concatenation and bilinear pooling and conduct comparative experiments on three stocks with different trends, the three stocks and their training and testing time ranges are consistent with those in Section 5.1. The experimental results are shown in Table 14.

From the comparison of experimental results, we can see that for three stocks with different trends, compared with the other two multi-type data fusion strategies, the multi-type data fusion method used in the MSF-DRL framework is more conducive to learning the optimal trading strategy, and can achieve higher TP, AR and SR.

5.5 Ablation experiments

In this section, to test the effectiveness of the multi-type data fusion MSF-DRL framework, we conduct an ablation experiment.

In the MSF-DRL framework, we fuse the stock data features, technical indicator features and image data features as the stock market state and take this as the input of RL. To validate the effect of the feature fusion of data with different structures, we conduct three groups of comparative experiments: one maintains the original MSF-DRL framework, the next set of experiments removes the analysis of candlestick charts in the MSF-DRL framework to implement algorithmic trading, and the last removes the analysis of stock data and technical indicators in the MSF-DRL framework to implement algorithmic trading. Note that in the ablation experiment, we choose the stock GE, the training time ranges from January 1, 2010, to January 1, 2018, and the test time ranges from January 1, 2018, to January 1, 2021. The comparison results are shown in Fig. 10 and Table 15.

The experimental results show that our trading strategy based on the analysis of image data or stock data and technical indicators can obtain profits under a downward trend of stock prices, nevertheless, the trading strategy learned based on the multi-type data fusion is significantly better than the other two trading strategies. Note that in the trading strategy learned under MSF-DRL framework, which can obtain profits based on the analysis of stock data, technical indicators and candlestick charts, all three of these components contribute to the learning of the trading

strategy, and the combination of the three makes the effect more obvious.

6 Conclusion and future work

In stock trading, some traditional trading strategies do not consider changes in the stock market, which often results in profits that are not ideal. An increasing number of studies take advantage of machine learning methods to implement algorithmic trading. The process of algorithmic trading using machine learning methods is affected by trader sentiment, and deep neural networks can obtain temporal characteristics from data with different structures. The environment of the stock market expression and optimal trading decisions are two major issues in algorithmic trading.

In this research, we combine deep learning and RL, and propose the MSF-DRL framework, which fuses the features of stock data, technical indicators and candlestick charts. The technical indicators are used to mitigate the impact of noise in stock data and to help capture major trends in the stock market. The candlestick charts are transformed by stock data and contain some information that is not present in the stock data. In the MSF-DRL framework, different neural networks extract the features of data sources with different structures, and a deeper feature representation of the stock market can be obtained by fusing data with different structures, which is more conducive to the learning of the optimal trading strategy.

From our experimental results, we can conclude that (1) the trading strategy based on the combination of deep learning and RL is better than the trading strategy based on single deep learning or RL; (2) stock data, technical indicators and the candlestick chart transformed from stock data can reflect the environment of the stock market; but the combination of the three has more obvious advantages and can reflect the environment of the stock market at a deeper level; and (3) compared with other trading strategies, our trading strategy exhibits performs better for stocks in different fields. In the overall comparison results, the average SR of our trading strategy is the highest, indicating that under the same risk, we obtain the highest profit.

The results demonstrate that our trading strategy can obtain better performance in the stock market. However, much textual information cannot be ignored, for example,

Table 15 Comparison results of ablation experiments on stock GE

	TP	AR(%)	SR	MDD(%)	MDDD
MSF-DRL(stock data and technical indicators)	13395.30	7.51	0.29	35.65	293
MSF-DRL(candlestick chart)	66813.24	21.63	0.61	53.88	209
MSF-DRL	311396.17	41.34	1.35	41.56	96

Bold entries signify the best results

some news events and investors' comments on the stock market, and there are many factors that affect stock price fluctuations. Therefore, it is important to extract informative data from the corresponding text for stock trading. In future research, we will consider using text information to obtain more diversified stock market information, to train a more stable and more robust trading strategy.

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Declarations

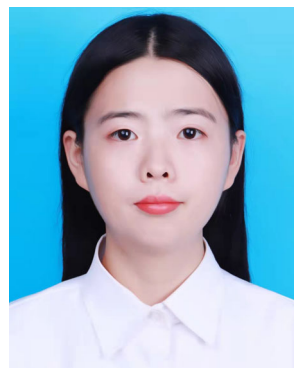
Conflict of Interests The authors declare that they have no conflict of interest.

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